LIVE

Polarisation

Unpolarized or Randomly polarized

⇒ If the plane of oscillation changes randomly for a transverse wave, then such a wave is called unpolarized or randomly polarized.



Bas 4 months

more

PR Prathik
y

RI Rishi Dalal
yes

PR Pranesh K.r
yes

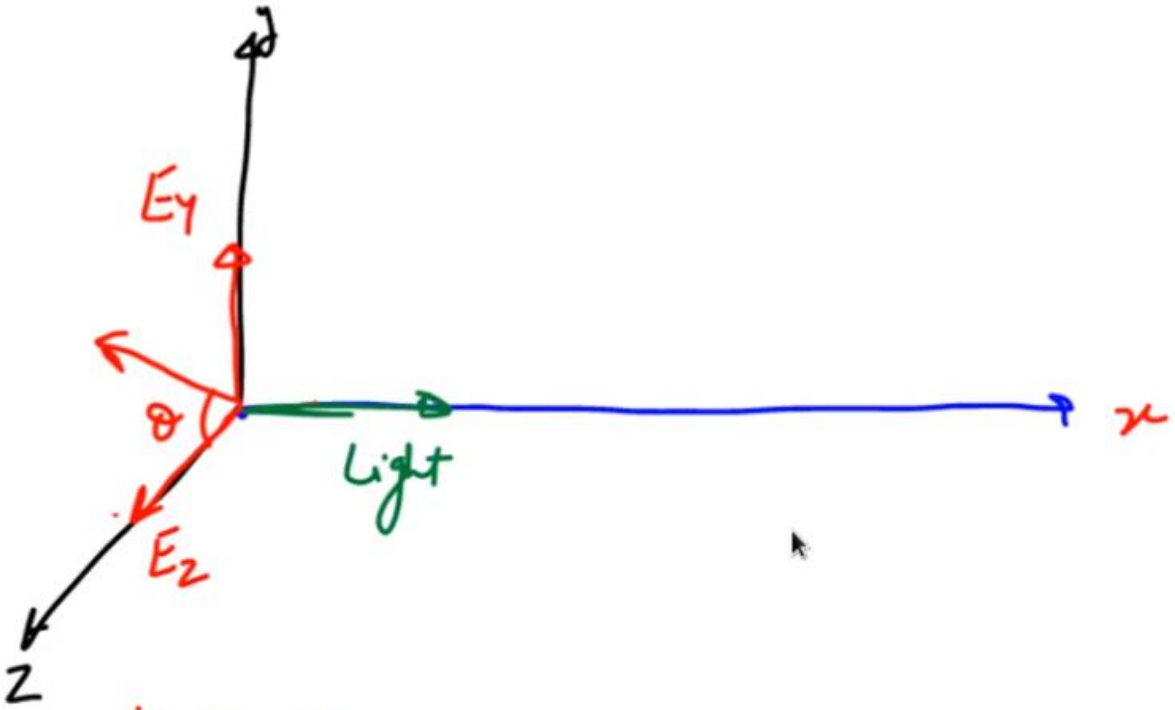
A A Hari kumar
y

HA Harsh Kanu
.

AV Aviraj Roy
yes

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$$\tan\theta = \frac{E_y}{E_z}$$

If θ changes randomly ⇒ unpolarized



yes

A Hari kumar
yes

Rishi Dalal
tanth

Jashan Grover insan
good morning sir

Aviraj Roy
yes

Sashi Raam v v
yes sir

Rishav Raj
.

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If θ becomes fixed $\Rightarrow$

such a light is called as linearly polarized or plane polarized

Plane of polarisation is the plane in which $\vec{E}$ & propagation of light exists



yes

RI Rishi Dalal
tanth

JA Jashan Grover insan
good morning sir

AV Aviraj Roy
yes

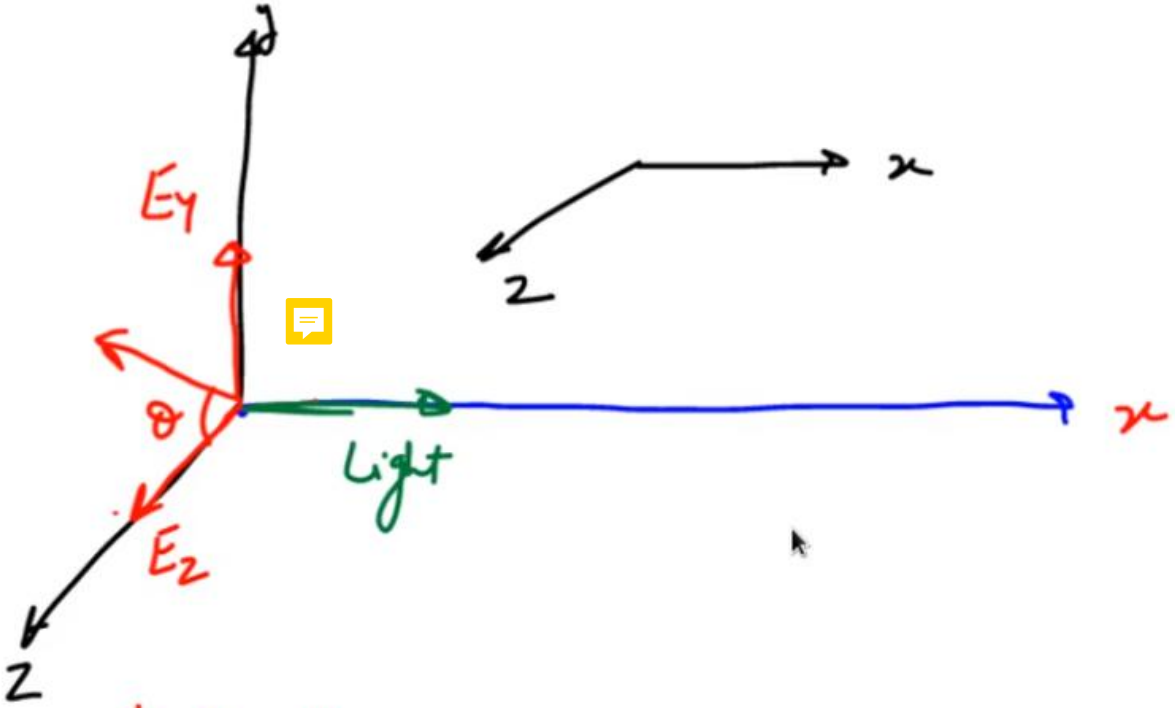
SA Sashi Raam v v
yes sir

RI Rishav Raj
.

RI Rishav Raj
polarised

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$$\tan\theta = \frac{E_y}{E_z}$$

If θ changes randomly ⇒ unpolarized

LIVE

tanth

JA Jashan Grover Insan
good morning sir

AV Aviraj Roy
yes

SA Sashi Raam v v
yes sir

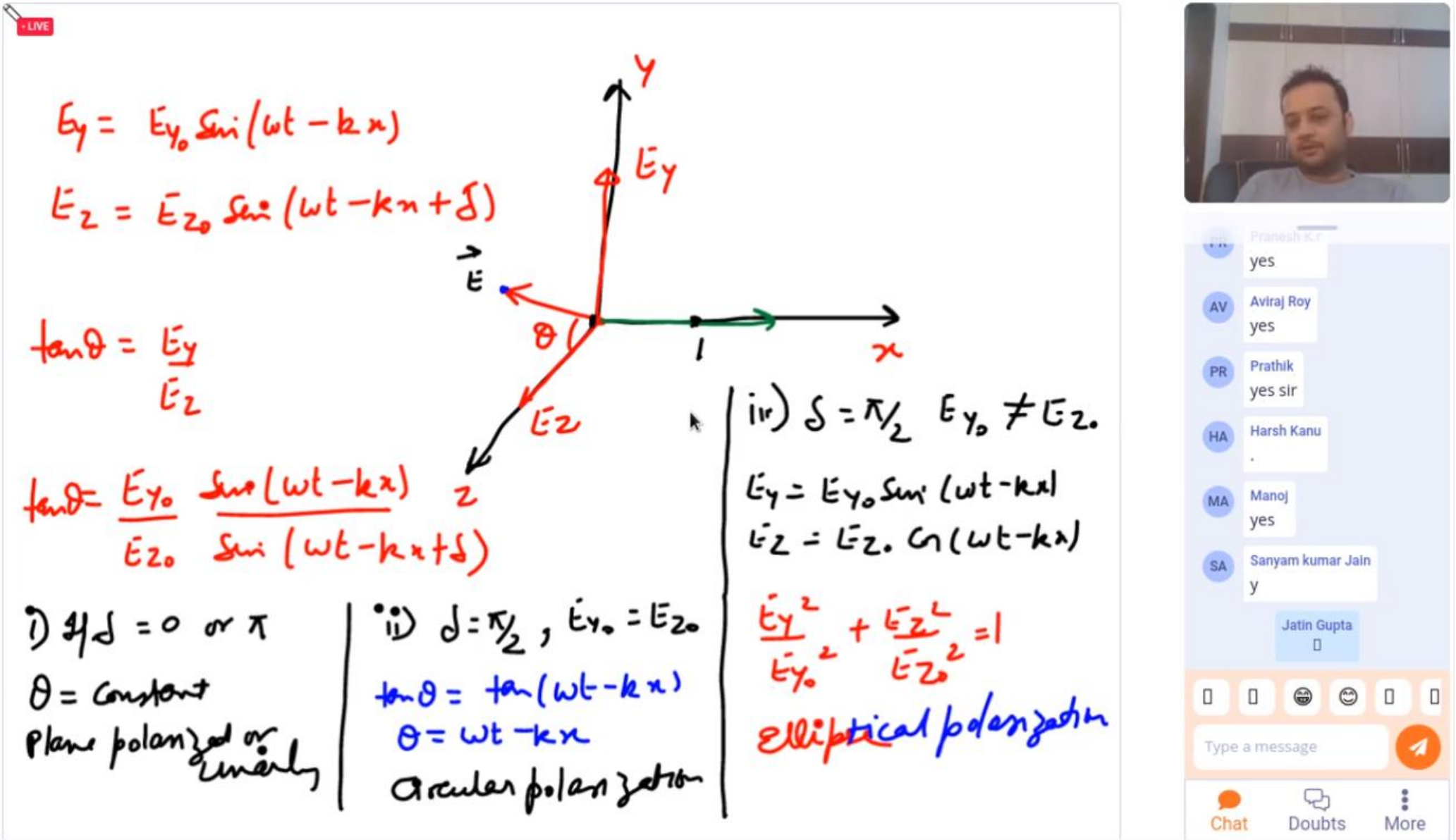
RI Rishav Raj
.

RI Rishav Raj
polarised

PR Prathik
yes

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Pranesh Kr
yes

AV Aviraj Roy
yes

PR Prathik
yes sir

HA Harsh Kanu
.

MA Manoj
yes

SA Sanyam kumar Jain
y

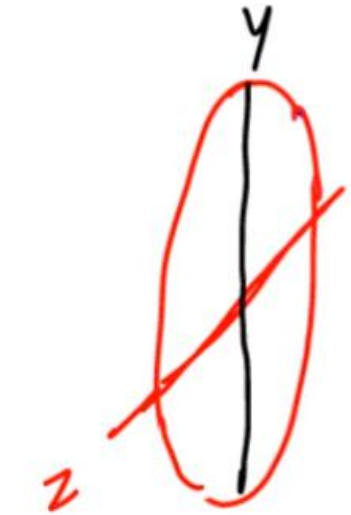
Jatin Gupta
▯

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LIVE

Plane polarization / Polaroids / Polarizers



(Transmission) Axis is Z

Polarizers have long chain of hydrocarbons which eliminate a certain fixed direction component of $\vec{E}$. The component ⊥ to this direction is allowed to pass. This ⊥ direction is also called transmission axis



Khushi
Yes

Jatin Gupta

PR Prathik
XZ

MA Madhav
Z

SA Sanyam kumar Jain
XZ

RI Rishi Dalal
ZX

PR Pratham omkar Pat...
XZ

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LIVE

Law of Malus

⇒ If a plane polarized light making an ∠ of θ with the transmission axis of a polarizer is allowed to pass through it, then the resulting intensity becomes $\cos^2\theta$ times



Rishi Dalal
zx

PR Pratham omkar Pat...
xz

RI Rishi Dalal
malu

A Hari kumar
😄

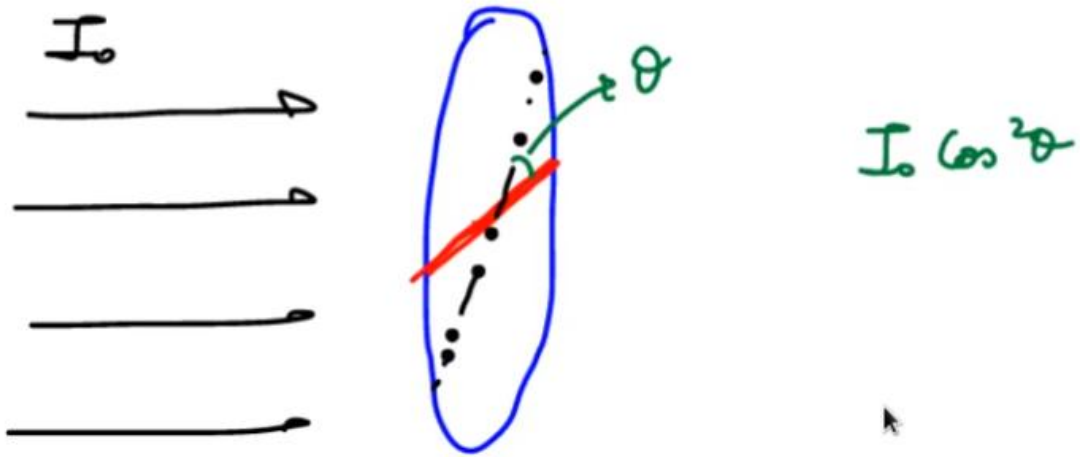
AV Aviraj Roy
*Law of Aloos, jokin

VA Vasu Khanna
xd

Aviraj Roy
😄

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$I_0 \cos^2\theta$



yes

RI Rishav Raj
.

PR Prathik
yy

HA Harsh Kanu
.

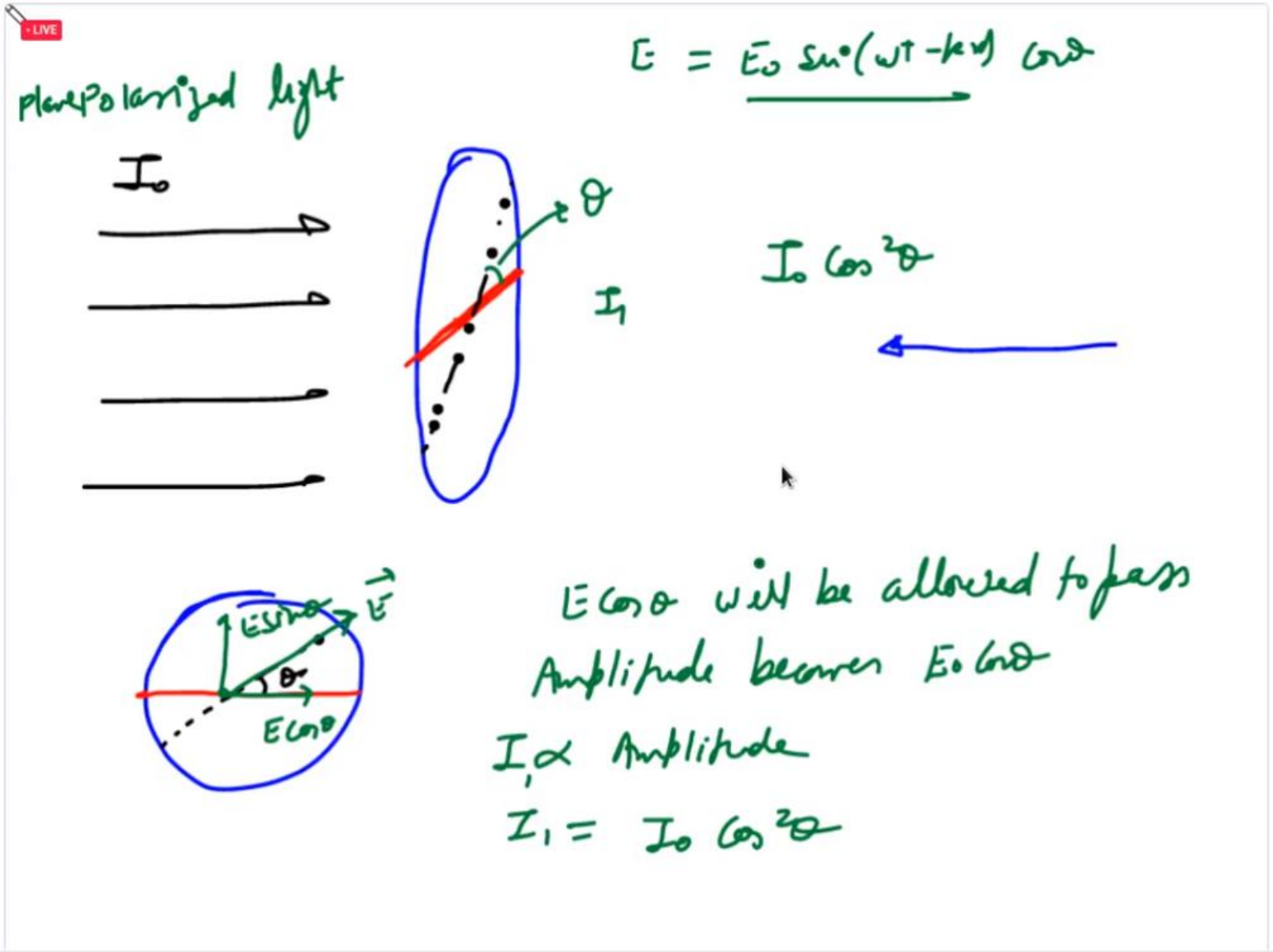
AV Aviraj Roy
yes

SA Sashi Raam v v
y

SR Srijahnavi Ch
y

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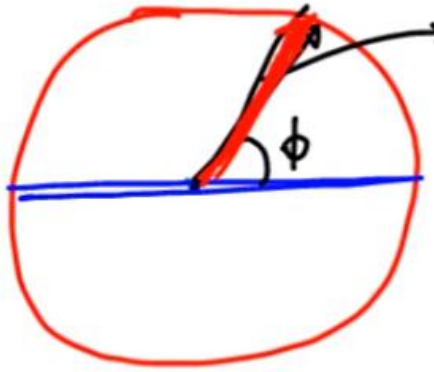

LIVE
plane polarized light
$E = E_0 \sin(\omega t - kx)$
I_0
θ
I_1
$I_0 \cos^2\theta$
$E\sin\theta$
$\vec{E}$
θ
$E\cos\theta$
$E\cos\theta$ will be allowed to pass
Amplitude becomes $E_0\cos\theta$
$I_1 \propto$ Amplitude
$I_1 = I_0 \cos^2\theta$



SA
Sashi Raam v v
yy
PR
Prathik
y
JA
Jatin Gupta
Yes sir
KE
Keshika Sharma
sir vo 2pi kaise ayega?
KE
Keshika Sharma
ha okay sir samjha
RI
Rishi Dalal
unitary meth
Type a message
Chat
Doubts
More

LIVE

⇒ If a randomly polarised light passes a polarizer then intensity becomes half



$$\int_{I}^{0} dI = \int_{0}^{2\pi} \frac{I_0}{2\pi}\, d\phi \cos^2\phi$$

$$I = \frac{I_0}{2\pi} \int_{0}^{2\pi} \cos^2\phi\, d\phi$$

$$= \frac{I_0}{2\pi} \times 4 \int_{0}^{\pi/2} \cos^2\phi\, d\phi$$

$$\boxed{I = \frac{I_0}{2}}$$



Sashi Raam v v: yy

Prathik: y

Jatin Gupta: Yes sir

Keshika Sharma: sir vo 2pi kaise ayega?

Keshika Sharma: ha okay sir samjha

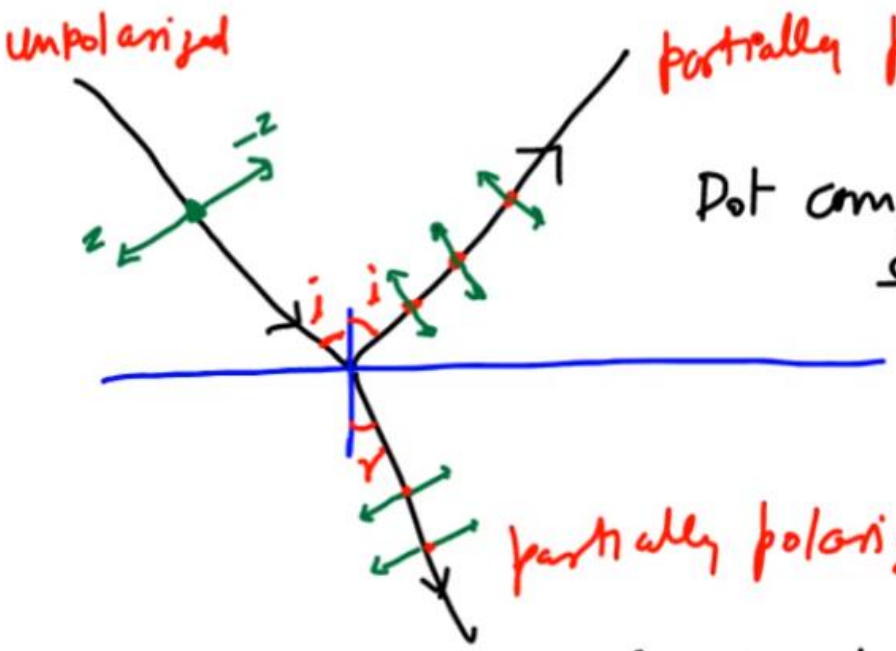
Rishi Dalal: unitary meth

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Chat Doubts More

LIVE

Brewster's law





PR Pranesh K.r
yes

A A Hari kumar
y

MA Madhav
.

RI Rishi Dalal
yes

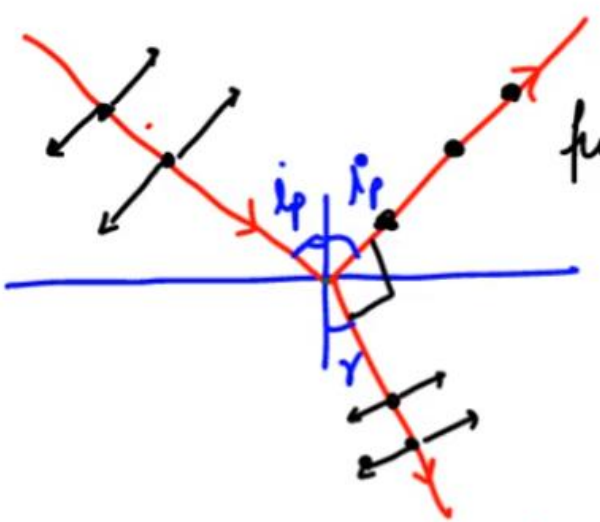
RI Rishav Raj
.

RI Rishi Dalal
ek jagah pe tendency bad gayi

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At a certain value of $i = i_p$, the reflected light is fully polarized. In this case reflected & refracted ray become $\perp$ to each other



fully polarized

$i_p + r = 90^\circ$

$r = 90^\circ - i_p$

$\mu_1 \sin i_p = \mu_2 \cos i_p$

$\tan i_p = \frac{\mu_2}{\mu_1}$

Brewster's $\angle$ or polarising $\angle$.

$i_p = \tan^{-1}\left(\frac{\mu_2}{\mu_1}\right)$



RI Rishi Dalal
ek jagah pe tendency bad gayi

SR Srijahnavi Ch
can u tell which is fully polarised, the arrow was not seen

SR Srijahnavi Ch
ok sir

RI Rishav Raj
.

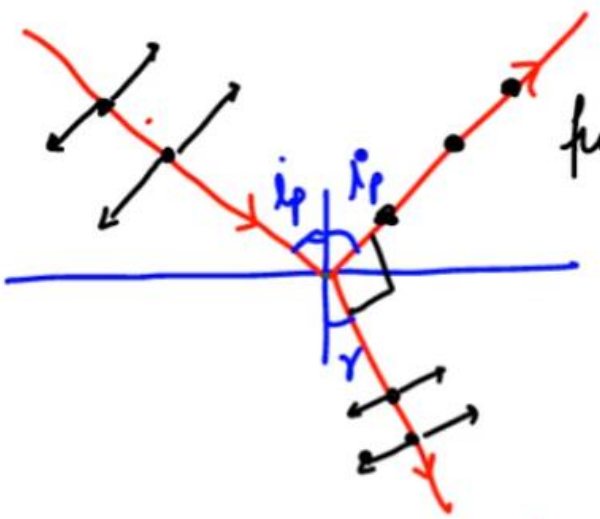
RI Rishi Dalal
g

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LIVE

At a certain value of $i = i_p$, the reflected light is fully polarized. In this case reflected & refracted ray become $\perp$ to each other



fully polarized

$i_p + r = 90^\circ$

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$\tan i_p = \frac{\mu_2}{\mu_1}$

Brewster's $\angle$ or polarising $\angle$.

$i_p = \tan^{-1}\left(\frac{\mu_2}{\mu_1}\right)$



RI Rishi Dalal
g

KH Khushi
Yes

SA Sashi Raam v v
yy

JA Jashan Grover Insan
sir how we concluded that two rays are perpendicular in this case ?

Harshal Sethiya

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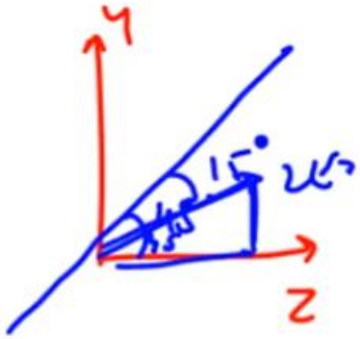
Linearly polarised light travelling along x-axis passes through a polariser sheet in y-z plane having transmission axis y = z. If electric field oscillations in incident light is represented by $E_y = E_0 \sin(kx - \omega t)$ and $E_z = \sqrt{3}E_0 \sin(kx - \omega t)$ then amplitude of electric field in transmitted light will be: [2016]

(a) $2E_0 \cos 15°$ (b) $E_0 \cos 45°$

(c) $2E_0$ (d) $E_0 \cos 15°$

$$\frac{E_y}{E_z} = \frac{1}{\sqrt{3}} = \tan 30$$

$$E^2 = E_0^2 + 3E_0^2 \Rightarrow E = 2E_0$$





Prathik
A

SA Sashi Raam v v
A

HA Harsh Kanu
a

RI Rishav Raj
2e

RI Rishi Dalal
2eo

AR Arnab Goyal
2E

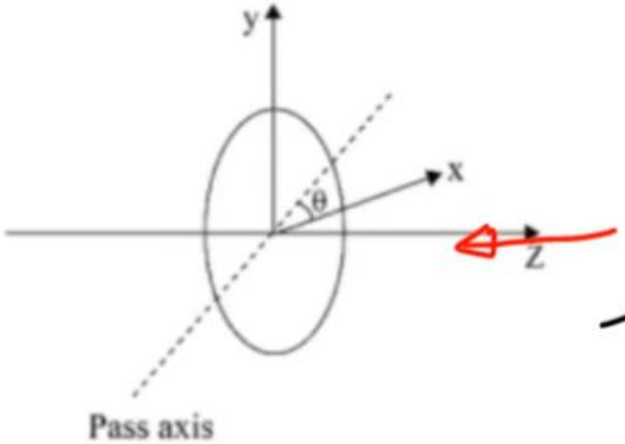
PR Prathik
2E0

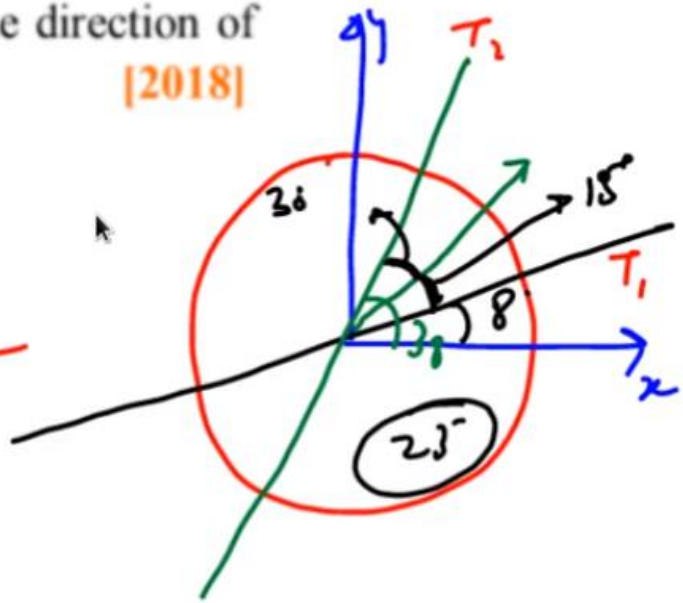
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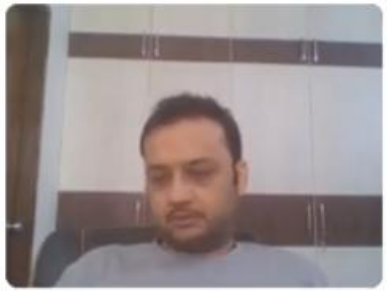
LIVE

A plane polarized light is incident on a polariser with its pass axis making angle θ with x-axis, as shown in the figure. At four different values of θ, $\theta = 8°, 38°, 188°$ and $218°$ the observed intensities are same. What is the angle between the direction of polarization and x-axis? [2018]





(a) 98°
(b) 128°
(c) 203°
(d) 45°



Jashan Grover Insan
38 or 21 b same

SA Sanyam kumar Jain
y

PR Prathik
y

AV Aviraj Roy
yes

RI Rishav Raj
haa comp of E along T same chahiye

JA Jashan Grover Insan
23 degree with x axis

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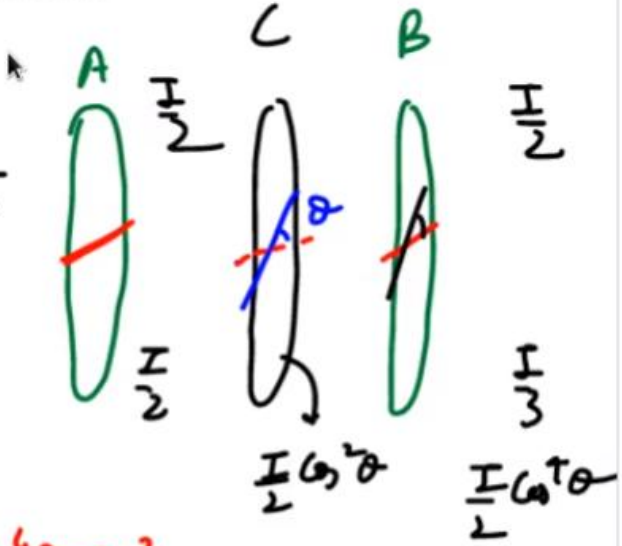
Unpolarized light of intensity I is incident on a system of two polarizers, A followed by B. The intensity of emergent light is $\frac{I}{2}$. If a third polarizer C is placed between A and B, the intensity of emergent light is reduced to $\frac{I}{3}$. The angle between the polarizers A and C is θ. Then:

(a) $\cos\theta = \left(\frac{2}{3}\right)^{\frac{1}{2}}$

(b) $\cos\theta = \left(\frac{2}{3}\right)^{\frac{1}{4}}$

(c) $\cos\theta = \left(\frac{1}{3}\right)^{\frac{1}{2}}$

(d) $\cos\theta = \left(\frac{1}{3}\right)^{\frac{1}{4}}$



$\frac{I}{2}\cos^4\theta = \frac{I}{3}$ $\quad \cos^4\theta = \frac{2}{3}$



Rishav Raj
.
A Hari kumar
yes
Jashan Grover Insan
again theta
Prathik
y
A Hari kumar
ye
Rishav Raj
y
Aviraj Roy
yes

Type a message

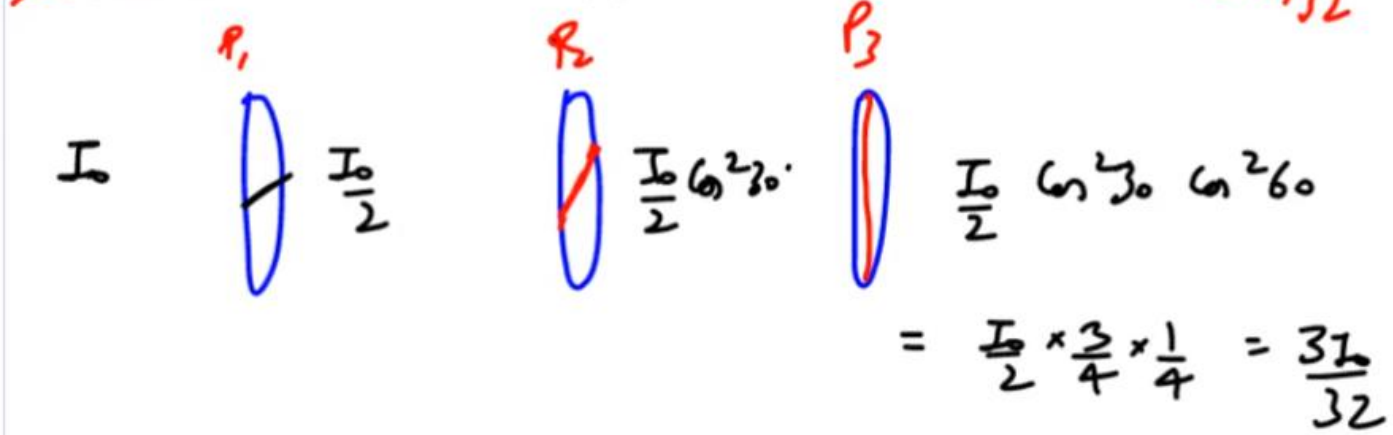
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A system of three polarizers P_1, P_2, P_3 is set up such that the pass axis of P_3 is crossed with respect to that of P_1. The pass axis of P_2 is inclined at 60° to the pass axis of P_3. When a beam of unpolarized light of intensity I_0 is incident on P_1, the intensity of light transmitted by the three polarizers is I. The ratio equals (nearly): [2019]

(a) 5.33 (b) 16.00

(c) 10.67 (d) 1.80





Sanyam Kumar Jain
30

KA Kartik Kumar
60

KA Kartik Kumar
oo sorry

A A Hari kumar
60

KA Kartik Kumar
yes

AR Arnab Goyal
yes sir

SA Sanyam kumar Jain
c

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LIVE

A beam of plane polarised light of large cross-sectional area and uniform intensity of $3.3 Wm^{-2}$ falls normally on a polariser (cross sectional area $3\times10^{-4}\ m^2$) which rotates about its axis with an angular speed of 31.4 rad/s. The energy of light passing through the polarizer per revolution, is close to: [2020]

(a) $1.0\times10^{-4}\ J$

(b) $1.0\times10^{-5}\ J$

(c) $5.0\times10^{-4}\ J$

(d) $1.5\times10^{-4}\ J$



ω = 31.4

ω = 10π rad/sec

N = 5

T = .2 sec

.1 × 3.1 × 3×10⁻⁴

9.9 × 10⁻⁵ = .000099

.0001

haa theta vary karega

Arnab Goyal: yes sir

A Hari kumar: T=0.2 seconds

Kartik Kumar: power

Rishi Dalal: Io/2 kaise sir? angle toh vary kar raha hai ryt?

Rishi Dalal: ha

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